

Daniel Gratzer

Assistant Professor, Aarhus University

Curriculum Vitae and Publication List

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Employment

- **Aarhus University** Tenure-Track Assistant Professor (Spring 2025 – Present)
- **University of Oxford** Visiting Postdoctoral Researcher (Summer 2024 – Winter 2025)
Hosted by **Sam Staton**
- **Aarhus University** Postdoctoral Researcher (Fall 2023 – Summer 2024)
Under **Lars Birkedal**

Education and Employment

- **Aarhus University** PhD (Fall 2018 – Fall 2023)
(Masters GPA: 12.0/12.0)
Thesis: *Syntax and semantics of modal type theory*
Advised by **Lars Birkedal**
- **Carnegie Mellon University** Bachelors of Science (Fall 2014 – Spring 2018)
(GPA: 3.73/4.00. Dean's List for 5 semesters. University Honors)
Honors Thesis: *The Next 700 Failed Step-Index-Free Logical Relations*
Supervised by **Karl Crary**

Research Visits

- **Chalmers University** with Thierry Coquand March 2022–June 2022
- **University of Paris** with Adrien Guatto December 2021
- **Aarhus University** with Lars Birkedal June 2017–August 2021

Publications

In my field conferences are the typical venue and a conference publication is full peer-reviewed research paper. While all conferences were ranked B or higher, several are rated at A* by the **Australian CORE rankings**. This signifies that they are in the top 4% of CS conferences. CORE rankings for journals ceased in 2020, but in the latest ranking two of the journal below were rated at A, signifying that they were within the top 10% of computer science journals.

23. **The ∞ -Category of ∞ -Categories in Simplicial Type Theory** 2026
Daniel Gratzer, Jonathan Weinberger, Ulrik Buchholtz
Logic in Computer Science (LICS)
CORE A* flagship conference
22. **Strict Universes for Grothendieck Topoi** 2026
Daniel Gratzer, Michael Shulman, Jonathan Sterling
Theory and Application of Categories (TAC)
21. **Normalization for multimodal dependent type theory** 2026
Daniel Gratzer
Logical Methods in Computer Science (LMCS)
Core A journal
(Extended version of *LICS paper by the same name*)
20. **Controlling Unfolding in Dependent Type Theory** 2025
Daniel Gratzer, Jonathan Sterling, Carlo Angiuli, Thierry Coquand, Lars Birkedal
To appear in *Mathematical Structures in Computer Science (MSCS)*
19. **The Yoneda embedding in simplicial type theory** 2025
Daniel Gratzer, Ulrik Buchholtz, Jonathan Weinberger
Logic in Computer Science (LICS)
CORE A* flagship conference
Distinguished paper
18. **Idempotent resources in separation logic: the heart of core in Iris** 2025
Daniel Gratzer, Mathias Møller, Lars Birkedal

17. **A modal deconstruction of Löb induction** 2025
Daniel Gratzer
Principles of Programming Languages (POPL)
CORE A* flagship conference
16. **Unifying cubical and multimodal type theory** 2024
Frederik Aagaard, Magnus Kristensen, Daniel Gratzer, Lars Birkedal
Logical Methods in Computer Science (LMCS)
CORE A journal
15. **The Category of Iterative Sets in Homotopy Type Theory and Univalent Foundations** 2024
Daniel Gratzer, Håkon Gylterud, Anders Mörtberg, Elisabeth Stenholm
Mathematical Structures in Computer Science (MSCS)
14. **Towards univalent reference types** 2024
Jonathan Sterling, Daniel Gratzer, Lars Birkedal
Computer Science Logic 2024
13. **Strict universes for Grothendieck topoi** 2023
Daniel Gratzer, Michael Shulman, Jonathan Sterling
To appear in Theory and Applications of Categories
12. **mitten: a flexible multimodal proof assistant** 2023
Philipp Stassen, Daniel Gratzer, Lars Birkedal
TYPES 2022 Post-proceedings
11. **Under lock and key: a proof system for a multimodal logic** 2023
G.A. Kavvos, Daniel Gratzer
Bulletin of Symbolic Logic
10. **A stratified approach to Löb induction** 2022
Daniel Gratzer, Lars Birkedal
Formal Structures in Computation and Deduction (FSCD)
9. **Normalization for Multimodal Type Theory** 2022
Daniel Gratzer
Logic in Computer Science (LICS)
CORE A* flagship conference
Selected for extended version (181 submitted, 63 accepted, ~10 extended versions)
8. **A Cubical Language for Bishop Sets** 2022
Jonathan Sterling, Carlo Angiuli, Daniel Gratzer
Logical Methods in Computer Science (LMCS)
CORE A journal
7. **Modalities and Parametric Adjoints** 2022
Daniel Gratzer, Evan Cavallo, G.A. Kavvos, Adrien Guatto, Lars Birkedal
Transactions on Computational Logic (ToCL)
CORE A journal
6. **Multimodal Dependent Type Theory** 2021
Daniel Gratzer, G.A. Kavvos, Andreas Nuyts, Lars Birkedal.
Logical Methods in Computer Science (LMCS)
CORE A journal
5. **Transfinite Iris: Resolving an Existential Dilemma of Step-Indexed Separation Logic** 2021
Simon Spies, Leonard Gäher, Daniel Gratzer, Joseph Tassarotti
Robbert Krebbers, Derek Dreyer, Lars Birkedal
Programming Language Design and Implementation (PLDI)
CORE A* flagship conference

4. **Multimodal Dependent Type Theory** 2020
 Daniel Gratzer, G.A. Kavvos, Andreas Nuyts, Lars Birkedal.
Logic in Computer Science (LICS)
 CORE A* flagship conference
 Selected for extended version (174 submitted, 69 accepted, ~10 extended versions)
3. **Implementing a Modal Dependent Type Theory** 2019
 Daniel Gratzer, Jonathan Sterling, Lars Birkedal.
International Conference on Functional Programming (ICFP)
 CORE A* flagship conference
 Distinguished paper (119 submissions, 39 accepted and 4 distinguished papers).
2. **Cubical Syntax for Reflection-Free Extensional Equality** 2019
 Jonathan Sterling, Carlo Angiuli, Daniel Gratzer.
Formal Structures in Computation and Deduction (FSCD)
 Best Junior Researcher Paper
1. **Managing Obligations in Higher-Order Concurrent Separation Logic** 2019
 Aleš Bizjak, Daniel Gratzer, Robbert Krebbers, Lars Birkedal.
Principles of Programming Languages (POPL)
 CORE A* flagship conference

Selected Talks

I have grouped talks by the research they present. Only invited talks—as opposed to conference talks—are included.

- **The Category of Categories in Simplicial Type Theory** 2026
 - TYPES
- **The Yoneda Embedding in Simplicial Type Theory** 2024
 - University of Birmingham, Midlands Logic Seminar
- **Directed Univalence in Simplicial Type Theory**
 - University of Manchester, YAMCATS 2025
 - University of Oxford, OASIS Seminar 2025
 - University of Birmingham, Theory Seminar 2024
- **Syntax and semantics of modal type theory**
 - University of Cambridge 2025
 - University of Oxford, PUDDLE Seminar 2024
 - University of Lyon, Chocola Seminar 2024
 - Homotopy Type Theory and Computation 2024
 - Homotopy Type Theory Electronic Seminar Talks (HoTTEST) 2023
 - Bergen University, Type Theory Seminar 2023
- **Controlling unfolding in type theory** 2023
 - Second WG6 meeting
- **Normalization for multimodal type theory** 2022
 - Chalmers University, ProgLog Seminar
- **Sketching type theories** 2021
 - University of Lyon, Chocola Seminar
- **Multimodal Dependent Type Theory**
 - Stockholm University, Logic Seminar 2020
 - IT University of Copenhagen, PLZ Seminar 2020
 - University of Paris 2019
- **Implementing a Modal Dependent Type Theory** 2019
 - Computer Science Day in Aarhus University

Teaching and Service

- **Organizer** for HoTT/UF (2025, 2026)
- **Program committee member** for MFPS'22, HoTT/UF'24, LICS'25, ICFP'25, TYPES'25, FSCD'26
- **Layout Editor** for LMCS (Spring 2019 – Spring 2021)
- **External reviewer** for CSL, ESOP, POPL, ITP, ICFP, JFP, LICS, and MSCS.
- **Lecture Series**
 - Foundations of Guarded Recursion at EPIT Summer School (Spring 2025)
 - Dependent Type Theory at University of Oxford (2024)
- **Teaching at Aarhus University**
 - Lecturer for Computability and Logic (Spring 2026)
 - Lecturer for Category Theory (Spring 2026)
 - Lecturer for Advanced Topics in Programming Language Theory (Fall 2025)
 - Lecturer for Modern Dependent Type Theory (Spring 2024)
 - Co-organizer for Categorical Type Theory Seminar (Spring 2021)
 - Co-organizer for Category Theory (Fall 2019, Fall 2020)
- **Supervision**
 - Rahul Chabra, PhD (joint with Lars Birkedal) 2025–present
 - Zoe Stafford, Summer Internship 2025
 - Julian Marquet, Masters Internship (joint with Lars Birkedal) 2025
 - Mathias Adams Møller, Research Assistant (joint with Lars Birkedal) 2024–2025